Laraib Mustafa Mirza

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Professional Profile

M.Sc. graduate in Astrophysics and Cosmology interested in the multi-wavelength emission of high-redshift blazars and the growth of their black holes, with hands-on experience in *NuSTAR*, *Swift* and *Fermi*-LAT data analysis, broadband SED construction, relativistic jet modelling (JetSeT) and Bayesian inference (MCMC).

Research Interests

High-Energy Astrophysics, Relativistic Jets and Active Galactic Nuclei, Growth of Supermassive Black Holes in the Early Universe, X-ray and γ-ray Astronomy, Multi-wavelength and Data-intensive Astronomy, Bayesian Inference in Astronomy

Education

M.Sc. in Astrophysics and Cosmology — Graduated Oct 2026
University of Padua, Italy

B.Sc. in Physics — 2021
Jamia Millia Islamia, New Delhi, India

Technical Expertise

X-ray, γ-ray & Astronomical Data Analysis: HEASoft, NUSTARDAS, CALDB, XSPEC (model comparison, F-test, χ^2 profiles, confidence contours), *NuSTAR*, *XMM-Newton* and *Swift* XRT/UVOT data, *Fermi*-LAT analysis (Fermitools, fermipy), DS9, aperture and differential photometry.
Statistics & Modelling: Bayesian MCMC (emcee), frequentist fitting (Minuit/iminuit), convergence diagnostics, posterior and degeneracy analysis; JetSeT (relativistic jet radiation modelling).
Programming: Proficient: Python (NumPy, SciPy, Astropy, Matplotlib); Intermediate: C++, FORTRAN, C, Linux shell scripting.
Practices: Reproducible Jupyter pipelines, Git version control, LaTeX.
Languages: English (Fluent), Hindi (Native), Urdu (Native), Italian (Intermediate B1).

Research Experience

Master's Thesis: Jet Physics of MG3 J163554+3629 and Its Impact on SMBH Evolution in the Early Universe — 2025–2026
University of Padua — Supervisor: Prof. Eugenio Bottacini — Co-supervisor: Dr. Andrea Tramacere (Univ. of Geneva)

- Used hard X-ray data to separate the synchrotron self-Compton (SSC) and external Compton (EC) emission of a $z = 3.65$ blazar, and its broadband SED to constrain the black hole and its growth.
- Reduced and analysed *NuSTAR* hard X-ray data (32.8 ks) with HEASoft/NUSTARDAS; diagnosed background contamination in one module and fitted spectra in XSPEC (absorbed power law, $\Gamma = 0.94 \pm 0.34$; a broken power law is not statistically required).
- Assembled a radio-to-γ-ray SED (GBT/VLA, WISE, *Gaia*, *Swift*-UVOT, *XMM-Newton*, *NuSTAR*, *Fermi*-LAT) with systematics, upper limits and EBL absorption.
- Fitted a one-zone leptonic jet model with external disk, BLR and torus photon fields in JetSeT, with Minuit and Bayesian MCMC (emcee), over a grid of 7 accretion efficiencies ($\eta = 0.057$–0.3, spin $a = 0$–0.996); $\chi^2 = 26.3$ for 27 d.o.f.
- Showed that *NuSTAR* breaks the SSC/EC degeneracy: SSC provides 60–74% of the jet emission at 10 keV, and EC produces the spectral hardening towards 40 keV. Derived $M_{\rm BH} \simeq 1.8 \times 10^9\, M_\odot$, Eddington ratio $\lambda \simeq 0.08$, and a particle-dominated jet more powerful than the accretion disk.
- Combined jet energetics with black-hole growth models: the present accretion rate cannot explain the mass, which requires an earlier phase of faster accretion at low radiative efficiency ($\eta \lesssim 0.1$); the jet power favours a rapid but not maximal spin ($a \approx 0.6$–0.8 for the fiducial jet composition).
  Repository: https://github.com/Mirza-751/MG3-J163554-3629-SED-Modelling

Independent Research: Uniform SED Modelling of High-Redshift Blazars 2026–Present

Self-directed, extending the thesis to the $z > 3$ *Fermi* blazar population

- Building a reproducible pipeline from public archival data to physical jet parameters (*Fermi*-LAT with Fermitools/fermipy, *Swift* and *NuSTAR* with HEASoft, SED fitting with JetSeT); first application to TXS 0800+618, with γ-ray light curves, multi-epoch SEDs and *NuSTAR* spectral fits.

Angulus: Spectrum-Centric GNSS Integrity Monitoring Nov 2025–Present

Work Package Lead: Simulation & Validation — Reference: Prof. Nicola Laurenti, University of Padua

Simulation and validation pipelines for frequency-based GNSS spoofing detection, in an interdisciplinary team.

Academic Projects

- **Supernova Natal Kicks of Compact Objects:** Simulated 10^4 massive stars in a Milky Way potential with `fireworks` + `SEVN`; compared the Hobbs kick model at solar and sub-solar metallicity; found neutron stars more widely distributed than black holes due to stronger natal kicks. Presented at MW-Gaia School 2023.
  Repository: https://github.com/Mirza-751/nbody-neutron-star-black-hole
- **X-ray Timing and Spectral Analysis of a Blazar:** Analysed 7 epochs of *Swift* XRT/UVOT data (2005–2022) of 1ES 2344+514; aperture photometry and XSPEC spectral modelling; identified flaring and long-term variability.
  Repository: https://github.com/Mirza-751/blazar-1ES2344
- **Exoplanet Transit Detection:** Reduced 1857 frames from the Asiago 1.82 m telescope (differential photometry, 0.0038 scatter); TESS analysis with Wotan + TLS; MCMC with BATMAN refining $P = 1.4391$ d, $R_p/R_s = 0.1270$, $i = 84.09°$ for Qatar-1b.
  Repository: https://github.com/Mirza-751/qatar1b-transit-analysis
- **Cosmological Perturbation Theory:** Derived perturbed Einstein equations in synchronous and conformal Newtonian gauges (Ma & Bertschinger); built gauge-invariant Bardeen potentials and connected them to CMB and matter power spectrum observables.
  Repository: https://github.com/Mirza-751/cosmological-perturbation-theory

Conferences & Workshops

Early Universe from Home Feb 2026

Virtual Conference — https://earlyuniverse.discussingresearch.com

X-Ray Spectroscopy Workshop Aug 2023

MIT, USA (Virtual) — https://cxc.harvard.edu/cdo/hrxs2023

MW-Gaia School Jul 2023

Vilnius University, Lithuania — https://mw-gaia.org

Presented: "Stellar Dynamics and Supernova Kick Simulations in the Milky Way"

JWST Training School Jun 2023

Instituto de Astrofísica de Canarias, Spain — https://meetings.iac.es/jwstworkshop2023

Gravitational Wave Data Workshop May 2023

University of Padova, Italy

Professional Experience

AI Training Specialist Dec 2025–Present

Alignerr and Mindrift (Remote)

Evaluation and alignment of AI reasoning on scientific tasks, including astrophysical plot interpretation.

Other Activities

- **Tutor Buddy for International Students**, University of Padua: supported incoming students.
- **Member**, COST Actions CA21126 (NanoSpace) and CA21106 (CosmicWISPers), 2023–Present.

References

Prof. Eugenio Bottacini
Thesis Supervisor, University of Padua
eugenio.bottacini@unipd.it

Dr. Andrea Tramacere
Co-supervisor, University of Geneva
andrea.tramacere@gmail.com